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A review of recent efforts in digitalization and application of endangered scripts

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Endangered scripts carry local knowledge and culture, making digitalization vital for cultural continuity. This paper reviews 120 publications (2011–2025) through computing and design lenses and proposes a phased classification. Our analysis shows that the literature prioritizes technical aspects of digital preservation, while application-oriented studies remain limited. This paper argues for a paradigm shift from preservation to revitalization, calling for community partnership, cultural grounding, and collaboration with creative disciplines.

Languages face an even greater risk of extinction than biological species¹. By the century's end, 50% to 90% of the world's 6900 languages are projected to disappear². Script endangerment is inextricably tied to this crisis. Colonialism, globalization, and the spread of colonial languages have left many Indigenous scripts vulnerable, disrupting intergenerational transmission, shrinking usage domains, and limiting digital infrastructure³. In Indigenous communities, language and writing are closely to land, ecological knowledge, and worldviews, but these connections have been eroded by oppression³. Not simply as tools for transcribing language, scripts are key media through which knowledge is organized and meaning articulated^{4,5}. Building on the 'Linguistic Relativity Hypothesis'^{6,7}, recent work has advanced the 'Script Relativity Hypothesis,' which argues that writing systems can also shape human patterns of thought⁸. Consequently, when a script disappears, both its diversity and culturally specific epistemologies and knowledge systems are lost. For instance, transliterating Aksara Bali into Latin characters may preserve phonetics but sever its ties to the ritual authority and palm-leaf manuscript's aesthetic traditions⁹. The cultural meanings embedded in scripts show that advances in digitalization are essential but not sufficient. Safeguarding should be translated into sustainable, practical support that enables communities to continue writing, teaching, and creative use while also strengthening public understanding and engagement. Accordingly, the importance of Indigenous languages and scripts, together with the crises they face, has prompted international action, such as UNESCO's International Decade of Indigenous Languages (IDIL 2022–2032) initiative (<https://www.unesco.org/en/decades/indigenous-languages>).

As digital technologies accelerate, 'Digital Heritage' emerges as an extension of traditional heritage revitalization paradigms¹⁰. A survey by Münster illustrates the multidisciplinary composition of researchers in the digital heritage field, consequently showing an interdisciplinary trend¹¹. Similarly, a comparable dynamic is observed in endangered script research, where emerging technologies, humanistic interpretations, and design-driven experiences offer new opportunities for preservation and cultural

innovation¹². Digitalization methods, such as character encoding, type design, input methods, and interactive experiences, enable endangered scripts to be both 'seen' and 'usable.' Moreover, the focus of digital heritage research and practice shifted from evaluating the effectiveness of tools to deeper social questions, concerning who produces these resources, for what purposes, how they are used, and by whom their legitimacy is defined¹³. A systematic examination of these questions helps clarify how digitalization practices influence the visibility, public approachability of endangered scripts.

Over the past five years, research on research on digitalization of endangered scripts has grown rapidly, with publication output increasing twelve-fold between 2011–2015 and 2021–2025 (Section 4.2). The development of technologies, such as Generative AI, has brought new opportunities to overcome long-standing limitations such as data scarcity, collection difficulties, and data pollution^{14–16}. Existing reviews have advanced understanding from multiple perspectives. For example, Girdhar et al. examined technical workflows for digitizing historical manuscripts¹⁷; Ajani et al. explored the role of digital technologies in Indigenous language learning¹⁸; Sihite and Sibarani summarized digital tools used in Indonesian language education¹⁹; Akallouch et al. reviewed advances in technologies for Amazigh²⁰; Taylor and Kochem discussed applications of computer-assisted language learning for revitalizing Indigenous knowledge²¹. These reviews tend to focus on particular languages or regions^{18–20}, specific technical pathways¹⁷, or pedagogical applications²¹, and scripts are typically treated as a subcomponent rather than the central object of inquiry. Few studies take endangered scripts as the primary unit of analysis to examine the global landscape of digitalization research. We aim to fill this gap.

This article presents the first systematic review that centers on endangered scripts. It examines the intersections of computing and design, maps the field's current landscape and distribution of contributions in terms of infrastructural feasibility and usability-oriented accessibility. To systematically review the current state of endangered scripts' digitalization, this paper proposes a three-stage classification framework: Alive, Active, and

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Applied, which we call *3A classification*. This framework is inspired by the digital heritage lifecycle²², which conceptualizes heritage work as a continuous process from creation to access and transformation. We analyze the literature in terms of research focus, technical approaches, and target users, and identify a structural pattern in current research: 58% of studies focus on foundational resource building (Alive), whereas only 18% reach community- and public-oriented applications (Applied), highlighting a substantial gap between technical development and translation of impact into social contexts.

Our contributions include:

1. The first literature review in the past fifteen years that centers on endangered scripts worldwide and examines their digitalization status, research focus, technological approaches, and target user groups.
2. The *3A Revitalization Pathway (Alive-Active-Applied)* proposed as a classification framework to systematically organize and review 120 relevant studies published between 2011 and 2025 and to map the field's distributional landscape.
3. Limitations and challenges identified and concrete insights offered to guide future work on endangered scripts' digitalization toward sustained technological innovation and stronger societal engagement.

Concept and Scope

This section establishes the conceptual grounding of the paper, focusing on endangered scripts, digital heritage, and their disciplinary intersections.

Endangered script. Scripts refer to a writing system consisting of a distinct set of visual symbols and structural rules, employed to record and transmit language²³. In contrast to language, which is a comprehensive system of sounds, vocabulary, and grammar, a script functions as its written form of expression⁵. The two are interrelated yet independent: a language can be recorded in multiple scripts (e.g., Turkish once used both Arabic and Latin alphabets), while a single script can also be shared by multiple languages (e.g., the Latin alphabet is adopted by hundreds of languages).

At present, there is no unified international standard for classifying endangered scripts. To support subsequent corpus inclusion and consistent eligibility decisions, and given the close linkage between scripts and languages, we adopt a multi-source identification strategy. We draw inspiration from the concepts in endangered-language research and its analytical approaches to risk dimensions and vitality factors, such as, UNESCO's five-level classification of language endangerment²⁴. We also consult the global documentation and visualization practices compiled in Tim Brookes' *Atlas of Endangered Alphabets* to supplement script-level identification cues (<https://www.endangeredalphabets.net/>). In addition, we consider the real-world contexts of endangered Indigenous scripts, including the impacts of colonial and postcolonial regimes³ and internal and external orthography disputes^{25,26} on social use, education, and intergenerational transmission. Building upon these studies, we define 'endangered scripts' as independent writing systems that, owing to social, political, or cultural factors, have lost their everyday communicative functions, survive only in restricted religious, ritual, artistic, or scholarly contexts, and face interrupted or imminently interrupted community transmission. Typical dimensions of assessment include:

- **Social usage:** A decline in the number of active users, with most community members shifting to dominant writing systems; the script ceases to serve core functions in education, administration, publishing, or everyday communication, persisting only in symbolic or ritual contexts.
- **Intergenerational transmission:** Younger generations lack functional literacy, and knowledge of the script is sustained only through a few cultural preservation initiatives or non-formal educational programs.
- **Digital ecosystem:** Absence of full encoding in standards such as Unicode, or only minimal implementation across operating systems,

input methods, fonts, and digital tools, rendering the script effectively 'invisible' or 'unusable' in digital environments.

This definition highlights the multidimensional nature of endangered scripts, reflecting both the decline of community use and the absence of technological and digital support. It provides operational criteria for subsequent literature screening and corpus construction.

Digital heritage and disciplinary perspectives. According to UNESCO's definition, 'digital heritage' includes "information created digitally" (born-digital heritage) as well as heritage "converted into digital form from existing analogue resources" (digitized heritage) (<https://unesdoc.unesco.org/ark:/48223/pf0000179529>). According to the characteristics of heritage objects, heritage can also be divided into two main domains: natural heritage and cultural heritage²⁷. Endangered scripts digitalization is part of digitized cultural heritage.

Contemporary digital heritage practice has moved beyond static archiving²⁸. In conservation science, the focus has expanded beyond documentation to include recording its geographic setting and historical context, so that material remains are preserved together with the circumstances²⁹. In work with Indigenous heritage, partnerships with Indigenous communities can be established through participatory action research (PAR)^{30–32} to jointly define problems, co-produce knowledge, and implement governance mechanisms. It reframes digitalization from a technical act of recording into a collaborative effort grounded in ethical legitimacy, sustainable impact, and reusable scholarly contributions.

These trends drive digital heritage research beyond disciplinary silos toward interdisciplinary collaboration. A survey illustrates the multidisciplinary composition of researchers in this field¹¹ (Fig. 1). Most studies come from the humanities (34%), with archaeology being the most prominent. This is followed by computer science (30%), geosciences (15%), architecture (8%), engineering (7%), natural sciences (4%), and design (2%). First, the 'Humanities' have made substantial contributions, and digital archaeology has emerged as one of the major focuses in digital heritage research in recent years²⁸. Social-scientific approaches^{31,32}, play an important role in fostering community-oriented collaboration by synthesizing historical trajectories, interpreting cultural contexts, and building chains of evidence. Second, the 'Computing' is treated as an independent disciplinary domain, highlighting the technologies and systems necessary for implementing digitalization and processing digital information, not equivalent to 'digital'. Third, 'Design' shows the lowest level of engagement at only 2%. Within the design discipline, systematic research and sustained engagement with digital heritage have yet to take shape. This limited involvement stands in tension with the field's need for approaches that support dynamic preservation and accessibility³³. Fourth, interdisciplinary collaboration has become essential, as no single discipline can address all aspects. A complete and representative digital heritage project typically involves coordinated contributions from experts across the social sciences, natural sciences, and art and design^{34,35}.

Therefore, in the context of digital heritage research, 'digital' does not merely mean computers or information technology itself, but refers to a whole research and practice framework centered on digital media and digital methods. The digital methods span the full spectrum of processes, including creation, storage, monitoring, dissemination, presentation, and preservation, aiming to safeguard the enduring value of heritage for long-term preservation, while ensuring accessibility, broad dissemination, and public engagement³⁶. This understanding resonates with the new discussed paradigm, where technology is treated as a tool, and information plays a coordinating role in establishing interdependent, symbiotic relationships among individuals, communities, environments, and tools³⁷. In this holistic approach, technological progress is no longer an isolated development, but rather a component deeply connected with humanity, society, and nature. This perspective shapes the methodological diversity of digital heritage, ranging from the interpretation of cultural significance in the humanities to the technical development in computing and user experience in design.

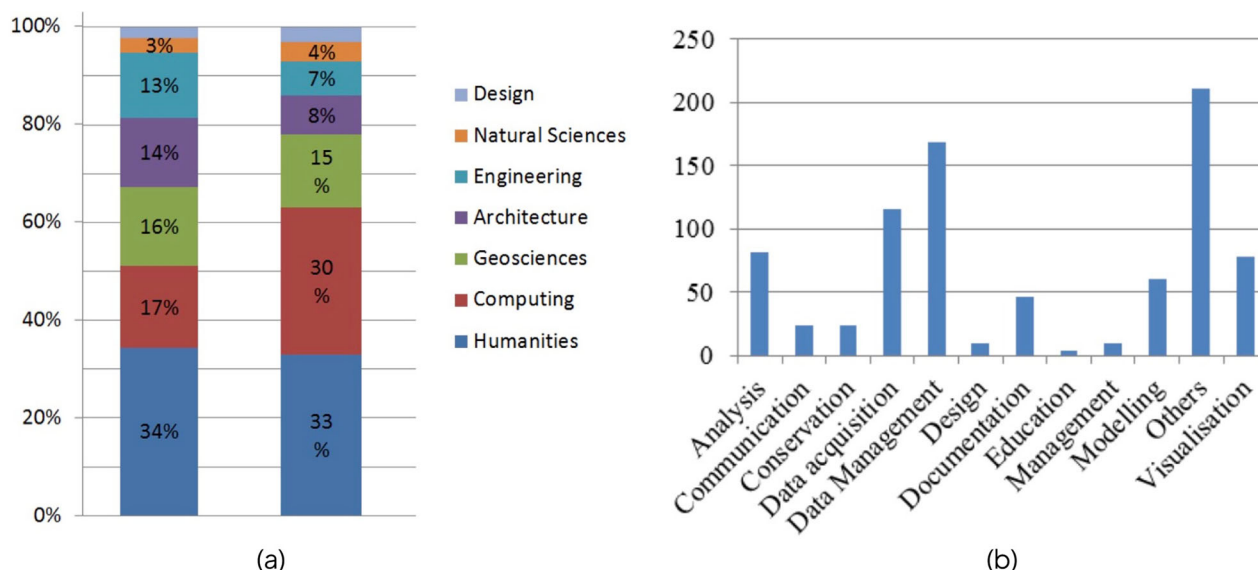


Fig. 1 | Disciplinary background and research topic from Münster's Survey¹¹. a Disciplinary background (left column: online survey, $n = 782$, right column: manual tagging of authors by institution, $n = 1232$) (b) topics. Licensed under CC BY 4.0.

Accordingly, in the specific research context of endangered scripts, digitalization is not merely a visual conversion of symbols. Intangible heritage (e.g., knowledge and culture) is deeply embedded in tangible media (e.g., manuscripts and inscriptions), the two form an inseparable whole^{38,39}. This symbiotic relationship implies that digitizing endangered scripts goes beyond remediating material artifacts and also involves integrating semantics and context to represent the cultural system as a whole.

Scope

As noted above, digital heritage practice is a collaborative process carried out by interdisciplinary teams. Within this process, social sciences contribute rich cultural evidence and key methodologies that connect technology with cultural meanings, helping ensure that digitalization remains grounded in its original cultural context and providing a foundational framework. This foundation is crucial, and its methods have been widely adopted in contemporary design disciplines, particularly HCI and design research. The designs have evolved into knowledge-producing paradigms, such as participatory design, user experience research, and service design. To maintain analytical depth and avoid an overly broad scope, this study focuses on computing implementations and design translation built on prior social-scientific research.

Recent studies indicate that digital heritage research spans multiple dimensions of heritage preservation and accessibility³³, where accessibility refers to open access, and also to how heritage knowledge is used and sustainably maintained. To support such accessibility, computing primarily provides foundational capacities such as encoding standards, font rendering, input methods, OCR, corpus processing, and platform applications, enabling scripts to be represented and processed in digital environments. Design, in turn, lowers barriers through interaction, presentation, and mechanisms of reuse, fostering understanding and participation. Digitizing endangered scripts faces a dual challenge of missing computational infrastructure and ruptured cultural contexts. The former determines whether a script can be represented and used in digital systems, while the latter dictates whether it can be understood, adopted, and sustained in practice. Accordingly, this paper centers on endangered writing systems. It approaches the current state of their digitalization through the lenses of Computing and Design (Fig. 2), providing a comprehensive review and discussion.

Material Collection

This review focuses on academic literature and adopts a cross-database, multi-keyword search strategy to capture interdisciplinary contributions from computer science, linguistics, the humanities, and design. Queries were conducted in October 2025 across Google Scholar, Web of Science, and the ACM Digital Library, covering the period 2011–2025. In terms of search strings, we included subject terms ("script" OR "written language" OR "character" OR "writing system"), condition terms ("endangered" OR "low-resource" OR "less-resourced" OR "minority" OR "Indigenous"), and process terms ("digitalization" OR "digitization" OR "digital preservation"). In addition to journal and conference papers, publication types covered workshop papers, posters, art gallery submissions, and master's and doctoral theses to capture diverse interdisciplinary outputs. The initial search identified 161 candidate publications (25 ACM, 43 WoS, 93 Google Scholar).

Following the initial search, 161 candidate publications were screened against predefined inclusion and exclusion criteria. **The inclusion criteria were:**

- Scripts with independent visual symbol systems and clear signs of endangerment (e.g., usage domain, transmission, digital infrastructure).
- Directly related to digital practices in computing or design discipline (e.g., encoding standards, font rendering, OCR, digital typeface design, interactive applications).
- Ancient scripts or historical manuscripts whose methods demonstrate relevance to endangered scripts digitalization.

The exclusion criteria included

- Scripts without independent visual systems, relying on external transcription schemes (e.g., IPA, Romanization).
- Endangered languages using non-endangered scripts (e.g., Kangri uses Devanagari with hundreds of millions of users)
- Computationally 'low-resource' scripts with strong social vitality (e.g., Urdu, Tamil, Amharic).
- Anthropological/policy studies unrelated to digitalization, Unicode proposals, non-English publications, preprints, or materials without accessible full texts.

Fig. 2 | Situated within digitized cultural heritage, the survey focuses on endangered scripts through two disciplinary perspectives: computing and design. The disciplinary categories shown within 'Digitized Cultural Heritage' are adopted from Münster's survey¹¹. The scope covers their bilateral intersections (Design $\cap$ Endangered Script, Computing $\cap$ Endangered Script) and their trilateral intersection (Design $\cap$ Computing $\cap$ Endangered Script).

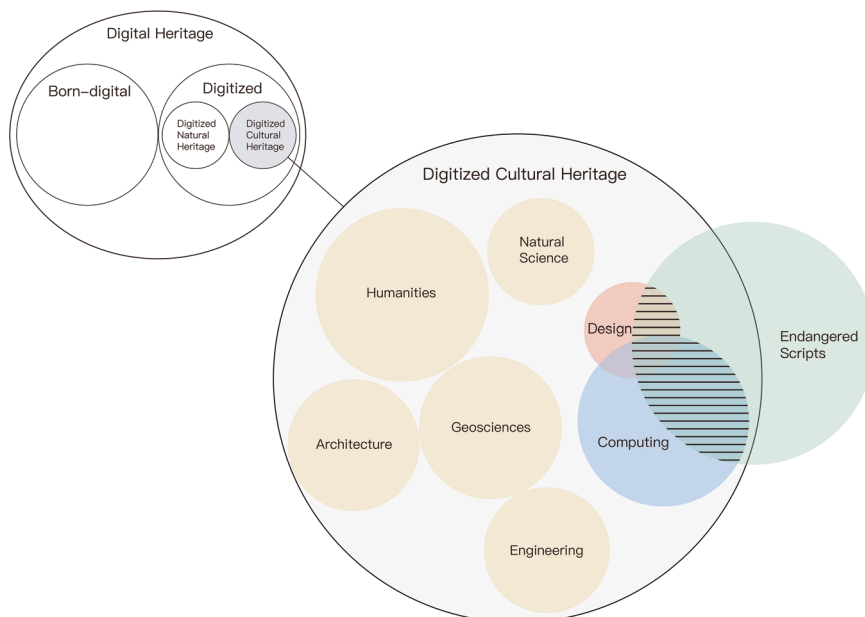
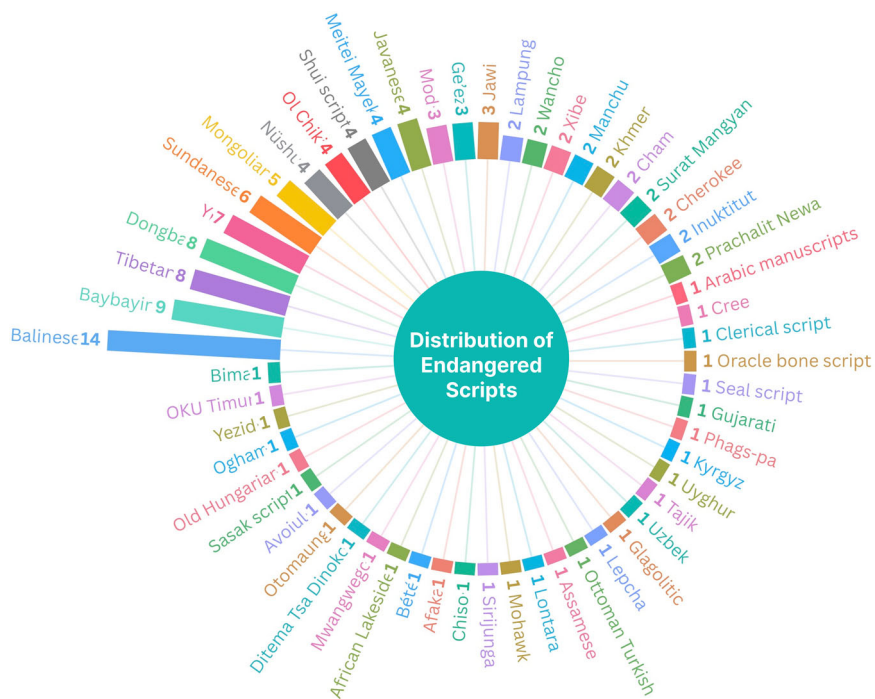


Fig. 3 | Frequency distribution of endangered scripts in 120 studies. The figure shows the number of times each endangered script appears in the reviewed literature.



This screening excluded 41 publications (10 under criterion one, 3 under criterion two, 25 under criterion three, 3 under criterion four), ultimately retaining 120 publications that directly address technological, cultural, and social challenges of endangered scripts digitalization. We coded the 120 included studies and assigned each a unique identifier, listed in the supplementary information.

Overall Patterns

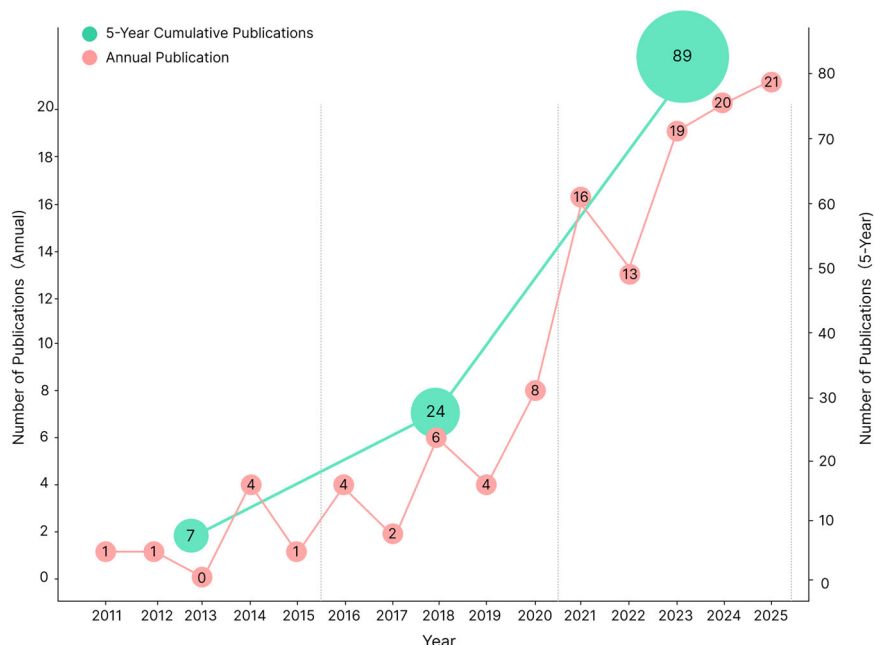
This section outlines the overall research landscape of endangered scripts' digitalization research by analyzing 120 publications, covering types of scripts, temporal trends, and geographical distribution. The results show that the types display a clear 'head-long tail' distribution, the research output shifted from fragmented to continuous growth; and spatially, the studies are

highly concentrated in Asia. Compared with the *Atlas of Endangered Alphabets* (<https://www.endangeredalphabets.net/>), overall patterns of resource density and research hotspots are broadly consistent, yet notable supply-demand mismatches emerge in Sub-Saharan Africa.

Types of scripts

Based on the 120 included studies, script entities were normalized and deduplicated by consolidating synonymous names and orthographic variants into unified identifiers. We then calculate the frequency of research attention across scripts and visualize the distribution (Fig. 3). The results reveal a clear 'head-long tail' pattern, in which a small number of scripts receive concentrated attention while the majority appear only once or twice, indicating that many endangered scripts remain significantly overlooked in

Fig. 4 | Temporal trend of research in endangered scripts’ digitalization (2011–2025). The red line indicates annual publications, and the green line shows 5-year cumulative totals.



current digitalization efforts. Leading the list are Austronesian and related traditional scripts from Indonesia and the Philippines, with Balinese receiving the highest frequency (14 mentions), followed by Baybayin (9), Tibetan (8), Dongba (8), Yi Script (7), Sundanese (6), and Mongolian (5). In stark contrast, about 51% of scripts appeared only once. This phenomenon drew our attention. Further analysis reveals that the distribution of research attention is shaped by academic and technical factors, as well as by the policy and legal environment. For instance, the Philippines established the legal status of Baybayin through *the National Writing System Act* in 2018⁴⁰, while Indonesia has promoted script revitalization initiatives through local regulations and activities (e.g., Balinese language month)⁴¹. These policy drivers have directly stimulated growth in related research, indicating a close correlation between ‘head’ scripts and institutional support.

Temporal trend

Figure 4 presents annual publication counts and five-year cumulative trends since 2011, showing rapid growth. During 2011–2015, annual output remained minimal (0–5 publications), reflecting exploratory pilot studies. Moderate growth occurred between 2016 and 2020 (cumulative total: 24), followed by a sharp rise after 2020, with annual output reaching 16 in 2021 and stabilizing around 20 in 2023–2025. Cumulatively, 2021–2025 produced 89 publications, almost twelve times the first five years. This growth stems from three factors: policy drivers, interdisciplinary trends, and technological development. Governments began to value cultural heritage digitalization issues. For instance, UNESCO launched the ‘International Decade of Indigenous Languages’ (IDIL 2022–2032) (<https://www.unesco.org/en/decades/indigenous-languages>) and the sub-project ‘The Missing Scripts’ (<https://sei.berkeley.edu/the-missing-scripts/>). Interdisciplinary research converges methods from computer science, linguistics, anthropology, and philology⁴², while advances in techniques (e.g., diffusion models, transformers, transfer learning) provide new possibilities for addressing data scarcity, material deterioration, and structural complexity^{43,44}.

Geographic distribution

By mapping scripts to their core usage areas (or historical origins) and grouping them into macro-regions, the spatial distribution shows clear geographic clustering, as illustrated in Fig. 5. The most active region is Southeast Asia, with high-density hotspots around Indonesia and the Philippines (e.g., Balinese, Baybayin, Sundanese, Javanese), totaling 47 instances (about 39%). Southwest China follows with 44 instances (36%), covering Yi,

Dongba, Nüshu, Shui, Tibetan, et al. South Asia forms a third cluster with 17 instances (14%), mainly from India and Nepal (e.g., Meitei Mayek, Ol Chiki, Modi). The remaining 13 instances (10%) are scattered across the Middle East, Africa, Europe, and the Americas (e.g., Ge’ez, Yezidi, Cherokee).

Comparing research hotspots (Fig. 5a) with resource points in Tim Brookes’ *Atlas of Endangered Alphabets* (Fig. 5b) (<https://www.endangeredalphabets.net/>), Asia’s dominance in endangered script digitalization aligns with its dense cultural-linguistic heritage and strong local communities. In contrast, other regions are more dispersed. It should be noted that some scripts span multiple regions through migration or shared history (e.g., Mongolian, Jawi). To avoid double-counting, each instance was assigned to a primary region based on community or origin, excluding cases with insufficient evidence from regional statistics. Even so, the global pattern of high concentration in Asia and sparse distribution elsewhere remains clearly visible. The comparison also reveals supply-demand gaps. Regions with abundant script resources, like Sub-Saharan Africa, have limited digitalization research, implying priorities for future data creation, community collaboration, and toolchain development.

A Multi-dimensional Analysis of Digitalization Practices

Following the overview, this section analyzes the literature in depth through three dimensions: research focus, technical pathways, and target users. We propose the *3A Revitalization Pathway* classification based on research focus and core contributions to categorize and analyze the literature. This section provides concrete outputs (such as models, systems, datasets, or applications) and analyzes 114 publications while excluding 6 reviews and conceptual papers^{17–21,45}.

Classification

To systematically analyze the research status of endangered script digitalization and reveal its evolutionary journey from digital preservation to cultural revitalization, this review adopts a phased perspective. We propose the *3A Revitalization Pathway* classification (Alive, Active, Applied) as a taxonomy, aiming to characterize the distribution of existing research through classification. The three stages correspond to the resource foundation, knowledge activation, and application empowerment, and reflect the common technical dependencies between tasks. They highlight the contributions of each study within the revitalization lifecycle, supporting a systematic comparison of goals and outcomes, rather than assuming a linear progression. For example, some studies may bypass early to produce direct

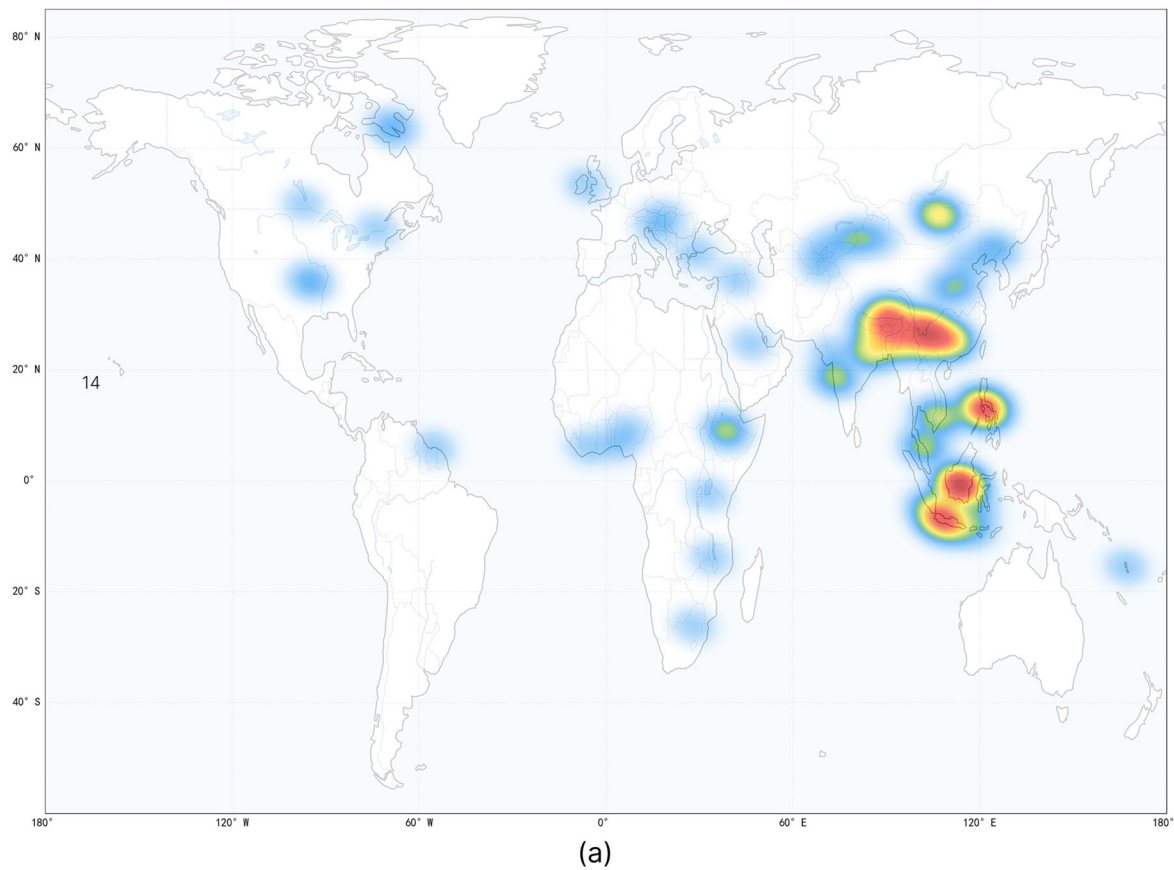


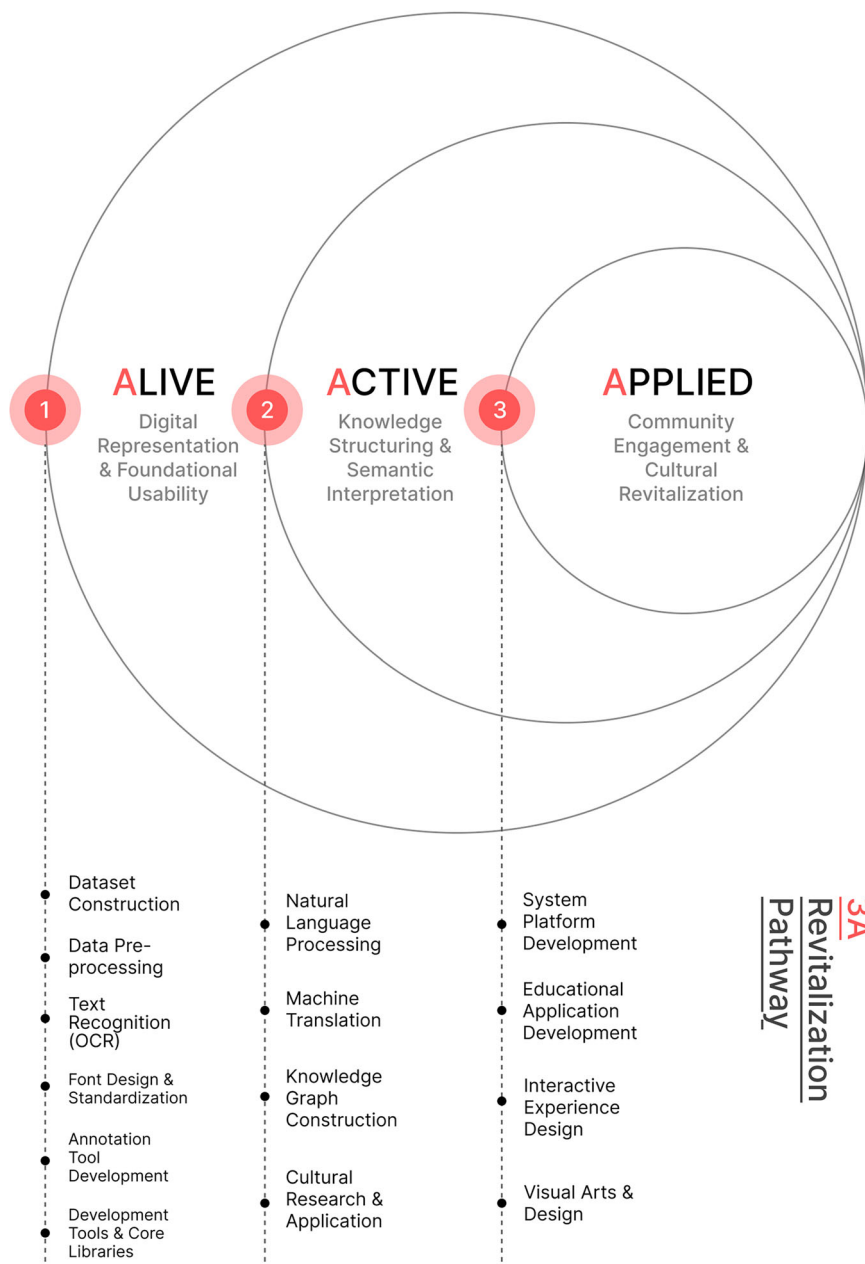
Fig. 5 | Geographic distribution of endangered script research and resources. **a** Spatial hotspots of research in endangered scripts' digitalization identified in this review. **b** Distribution of endangered script resources documented in *the Atlas of*

Endangered Alphabets (source: <https://www.endangeredalphabets.net/>), reproduced with permission from Tim Brookes.

public output when resources permit; or when community goals prioritize social use, high-impact public outputs may emerge even with limited technical investment.

Through iterative analysis of 114 empirical studies, we identified 14 thematic labels based on research objectives and core outputs. Two authors independently conducted open labeling on methodologies and objectives,

Fig. 6 | The 3A Revitalization Pathway classification. The framework outlines three stages of endangered scripts digitalization: Alive, Active, and Applied.



refining categories through constant comparison and discussion; a third author resolved discrepancies until the labels reached theoretical saturation. Studies with multiple aims were assigned primary and secondary themes, which were then mapped to the 3A classification framework (Fig. 6). These thematic labels demonstrate a shift from static digital archiving toward computational processing. Current research builds computable infrastructures (e.g., datasets, knowledge graphs, and reusable toolchains) and facilitates script reintegration into social contexts through education, interactive design, and public dissemination.

The classification analysis reveals a ‘foundation-first’ pattern. Alive dominates (58%, $n=67$), followed by Active (28%) and Applied (20%). The top three sub-objectives, ‘Text Recognition’ (39%), ‘Dataset Construction’ (18%), and ‘Natural Language Processing’ (13%) further confirm this focus on resource building. Notably, 20 studies exhibited multi-objective characteristics (Fig. 7), most frequently coupling ‘Dataset Construction’ and ‘Text Recognition’^{16,43,46–54}. These patterns highlight the strong technical coupling between foundational data and advanced computational processing.

We now turn to a detailed discussion of each stage in the 3A classification, focusing on their key contribution.

Stage I: ALIVE. The first stage is ‘ALIVE,’ with its core aim to give endangered scripts a digital identity. This stage primarily encompasses efforts that establish the foundational prerequisites for script computability, covering collection, processing, annotation, design, and development of basic recognition resources. By this criterion, any toolchain designed to facilitate resource production and standardization is also classified under the Alive stage. Research at this stage mainly covers six thematic labels: ‘Dataset Construction’, ‘Data Pre-processing’, ‘Text Recognition(OCR)’, ‘Annotation Tool Development’, ‘Font Design & Standardization’, and ‘Development Tools & Core Libraries’. Among the 114 studies reviewed in this research, those at the Alive stage account for the largest share (67 studies, 58%).

‘Dataset Construction’ serves as the essential foundation for endangered scripts’ digitalization, aiming to address the challenges of data scarcity, fragmentation, physical deterioration, and the high collection cost. Most

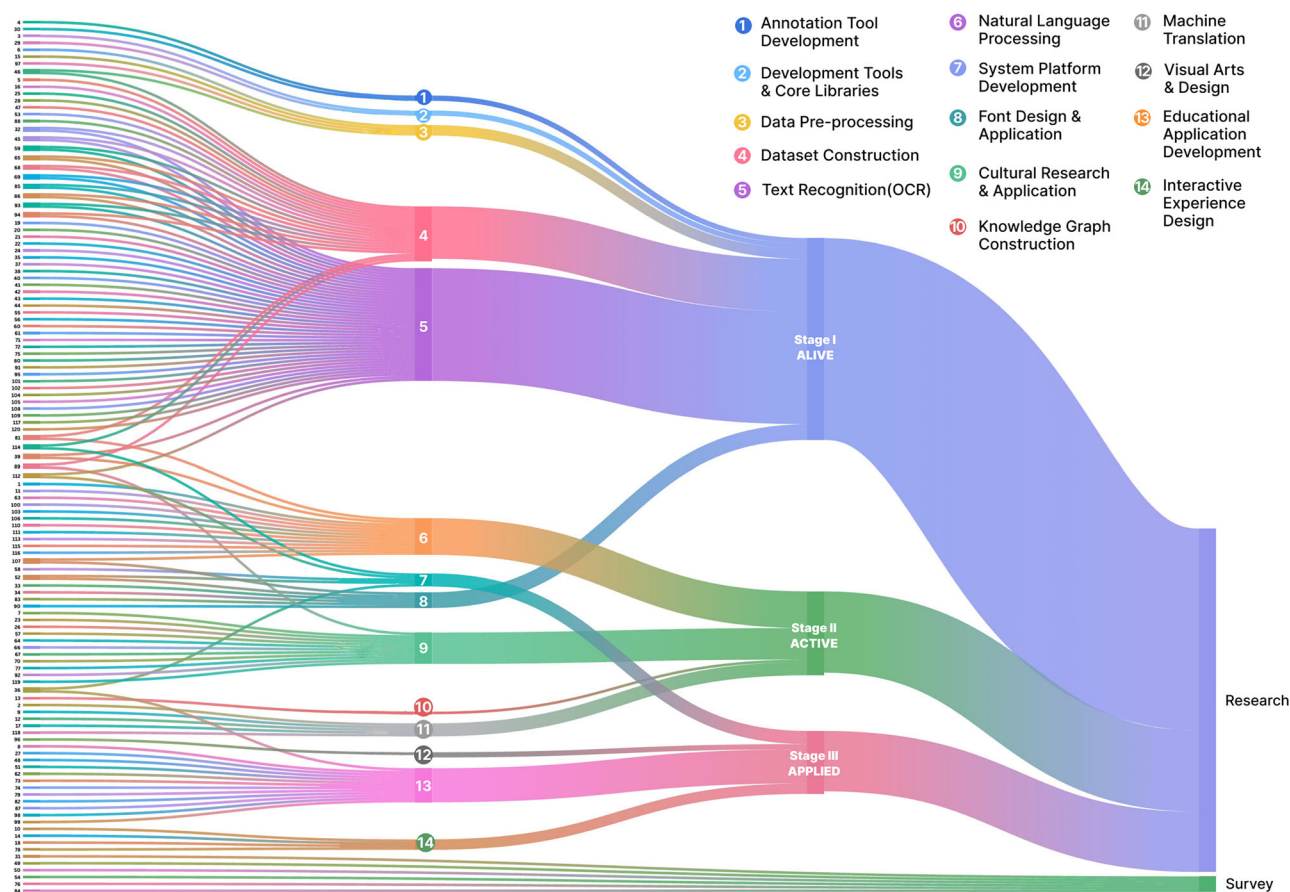


Fig. 7 | Sankey diagram of 114 studies under the 3A classification. The distribution of Alive, Active, Applied stages is visualized.

efforts reconstruct dispersed historical documents and manuscripts into standardized resources for computational research, providing reliable baselines for downstream tasks such as text recognition^{46,55}, as seen in the HHD-Ethiopic dataset⁴⁷ and the OG(H)AM project⁵⁶. While some studies capture contemporary writing practices through community-sourced samples⁵⁴, others bypass physical collection constraints (cost and scale) via algorithmic generation. Notable studies include noise reduction in generated imagery¹⁴, realistic background simulation¹⁵, and synthetic glyph augmentation¹⁶ to bolster recognition baselines.

‘Data Pre-processing’ optimizes the transition from degraded source material to reliable computational input by balancing quality and efficiency. Quality-focused research enhances signal readability^{15,57} and ensures annotation reliability⁵⁸. Studies for efficiency explore automation, clustering, and human-computer collaboration. To manage large-scale datasets, recent approaches prioritize reducing manual labor through preliminary structure extraction from unannotated data⁵⁹, freeing experts from repetitive processing for high-level validation and decision-making⁶⁰.

‘Text Recognition (OCR)’ represents Stage I’s core component (39% of sample), serving the key function of transforming visible traces into computable cultural resources. Research aims to achieve robust recognition under strict constraints such as limited training data and complex script morphology. Researchers made contributions in four areas. First, studies develop the first efficient and accurate recognition systems for specific endangered scripts, such as Sundanese⁶¹, Bima⁶², Surat Mangyan⁶³, Baybayin^{64–66}, Javanese⁶⁷, Modi⁶⁸, Gees⁶⁹, Pracalit Script⁷⁰. Second, researchers focus on script-specific challenges to ensure robust performance under limited data and high variability, including the visual complexity of Shui script^{71,72}, similar character forms in Manipuri⁷³, cursive structures in Manchu⁷⁴, segmentation in Tibetan⁷⁵, word-level recognition in Baybayin⁷⁶, boundary detection for the Dongba script⁷⁷ and feature-based recognition

using HSV color space⁷⁸. Third, studies provide transferable and comparable methodological advances, such as proposing more general frameworks^{44,79,80} or offering systematic evaluations and baseline references⁸¹. Fourth, studies aim to enhance automation and interoperability to bridge recognition with downstream applications. This linkage encompasses multi-script identification for processing mixed documents^{78,82}, automated normalization for NLP-ready outputs⁸³, and speech-text mapping to reconnect oral and written traditions⁸⁴.

To support large-scale and complex digitalization, three lines of research: ‘Annotation Tool Development’, ‘Font Design & Standardization’, and ‘Development Tools & Core Libraries’, focus on building essential infrastructure. **‘Annotation Tool Development’** establishes a scalable, reusable infrastructure to optimize data production by reducing costs and formalizing efficient workflows. While the process involves cultural interpretation, it outputs annotation resources and toolchains. Current research pursues two directions. One is community engagement, such as mobile crowdsourcing platforms that involve rural Indian communities to enhance efficiency while generating local income⁸⁵; and the other is approach optimization, exemplified by the DHelper toolkit, which integrates AI into Dunhuang manuscript studies to alleviate expert workload during the annotation phase⁶⁰. **‘Font Design & Standardization’** establishes a consistent digital presentation foundation by digital design and Unicode encoding⁸⁶. This is epitomized by The Missing Scripts project (<https://sei.berkeley.edu/the-missing-scripts/>), which has encoded over 120 scripts. Complementary work develops fonts and input methods for scripts like Cham, Balinese, and Santali, ensuring digital usability in everyday contexts^{87–90}. **‘Development Tools & Core Libraries’** provides long-term computability at the foundational level. Examples include Google’s Nisaba library for multi-script normalization⁹¹ and paradigm-

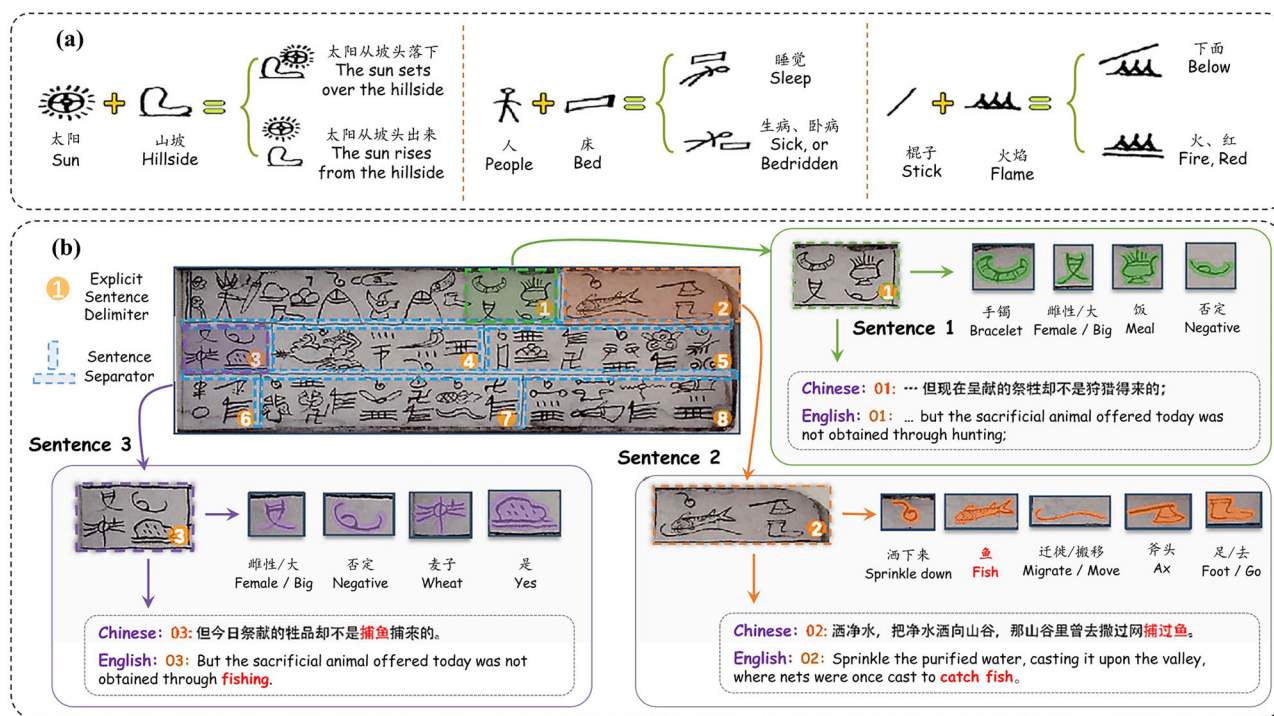


Fig. 8 | Illustration of Dongba script as a hybrid pictorial-linguistic system in Li et al.¹⁰⁸ **a** Spatial composition forms layered meanings through the visual arrangement of pictographs. **b** Example of paragraph-level segmentation showing sentence

grouping and contextual inference needed for accurate Dongba-Chinese translation. Licensed under CC BY-NC-ND 4.0.

shifting concepts for Indigenous programming systems that challenge colonial coding structures⁹².

Stage II: ACTIVE. Stage II (Active) (28%, $n = 33$) shifts focus from resource production to knowledge activation, imbuing scripts with semantic understandability, inferability, and translatability. Building upon established digital foundations, research in this stage integrates contextual expertise and social science frameworks to unlock deep structures and cultural values via computational methods. This stage encompasses four parts: 'Natural Language Processing,' 'Machine Translation,' 'Knowledge Graph Construction,' and 'Computational Humanities Research.' The primary outputs are semantic and interpretive structures that serve as the backbone for future educational and public platforms.

'**Natural Language Processing**' establishes linguistic operability for endangered scripts to ensure structured representation and cross-task transferability. A primary objective is overcoming barriers to daily digital utility, particularly through orthographic and transliteration solutions. For instance, a series of studies on Balinese script automated Latin-to-Balinese conversion by addressing derivational transformations⁹³, non-alphabetic symbols⁹⁴, orthographic conventions^{95,96}, and scriptio continua^{97,98}. Another effort preserves mathematical expressions in mobile environments⁹⁹. Other studies include linguistic modeling, such as capturing Tibetan morphological features¹⁰⁰ and Xibe syntactic parsing¹⁰¹. To facilitate knowledge discovery, research also implements automatic summarization¹⁰² and semantic metadata linking to enhance manuscript accessibility within global networks¹⁰³.

'**Knowledge Graph Construction**' elevates the outputs to semantic-level structuring, aiming to build cross-resource and computable knowledge support for endangered scripts through entity-relation diagrams. To address the core challenge of data sparsity, a representative study proposed combining structural information from graphs with textual descriptions from external encyclopedias¹⁰⁴. This approach improves knowledge representation and enhances the performance of downstream tasks such as reasoning and querying.

'**Machine Translation**' aims to establish semantic channels between endangered scripts and major languages, thereby improving their cross-context communicability. Existing studies address translation quality bottlenecks in low-resource settings, including challenges like handling special characters and low-frequency words in Mongolian-Chinese translation¹⁰⁵, and optimizing Tibetan translation with subword segmentation¹⁰⁶. Another study uses the structural similarities between Classical Chinese and the Dongba script to transfer linguistic knowledge and improve translation quality¹⁰⁷. Li et al. treat Dongba script as an iconic writing system, using multimodal features to enhance paragraph translation and resolve spatial semantic challenges (Fig. 8)¹⁰⁸. Further exploration attempts to integrate multiple tasks, such as developing an end-to-end system for Nüshu that performs recognition, segmentation, normalization, and translation in a unified workflow¹⁰⁹. These studies enable endangered scripts to move beyond local contexts and enter broader spaces of digital communication.

'**Computational Humanities Research**' extends technological outcomes to social and cultural dimensions. It aims to interpret the recontextualization and cultural functions of endangered scripts within contemporary digital environments, thereby providing the content organization, contextual explanation, and design rationale for future applications. Current research unfolds along three main directions. First, intra-textual analysis employs quantitative methods like text mining to examine thematic and historical changes¹¹⁰; another study constructs semantically indexed databases to investigate Dongba cultural and grammatical structures¹¹¹. Second, research investigates digital re-contextualization, exploring the symbolic potential of historical scripts in modern media^{112–114}. Some studies critically examine institutional barriers within socio-technical systems such as Unicode and font design frameworks^{115,116}. Third, digital ethnography analyzes script use and identity construction by investigating real-world writing practices on social media. This includes exploring user preferences for alternative scripts (e.g., Romanization)¹¹⁷, the role of online opinion leaders in language evolution¹¹⁸, and the cultural identity embedded in everyday literacy¹¹⁹. By synthesizing computational analysis with ethnographic/anthropological insights, these studies produce the essential

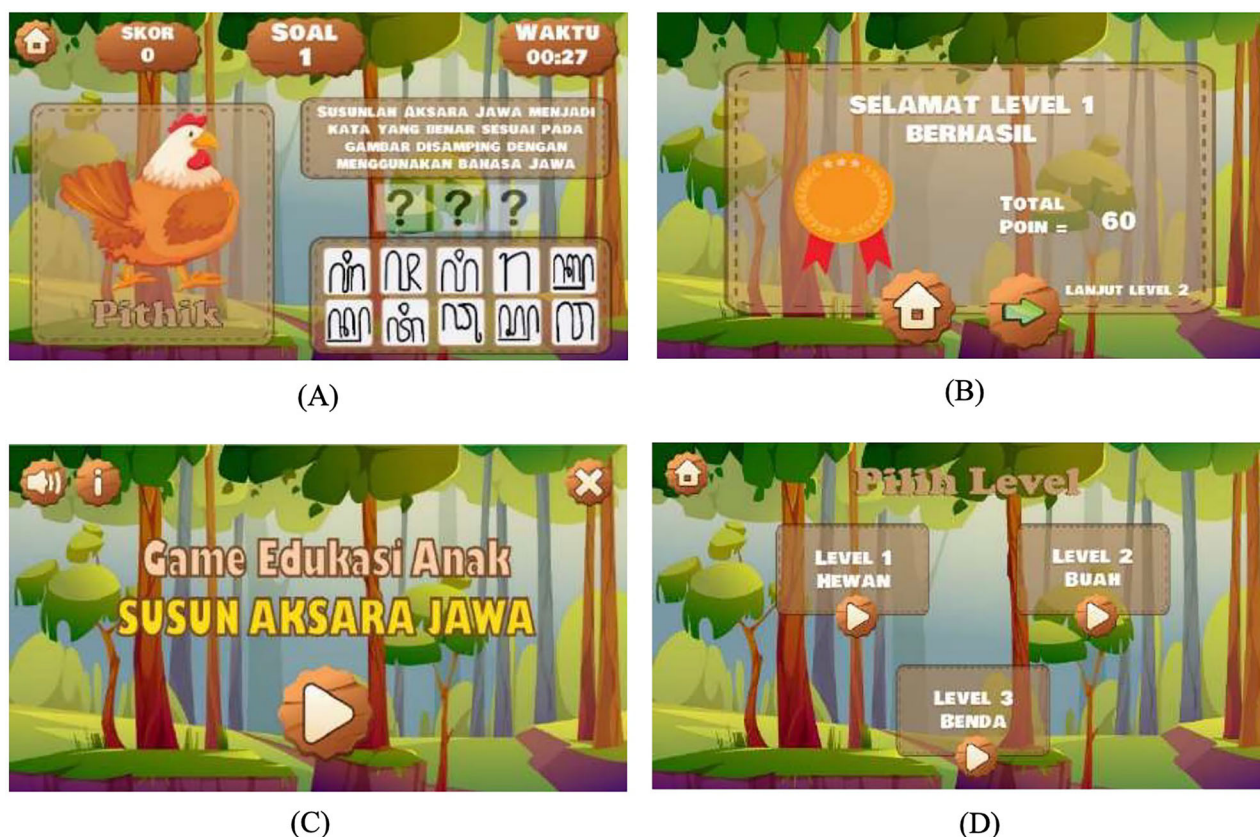


Fig. 9 | A gamified Javanese script learning application by Prasetyo et al.¹²³. **(A)** Main menu and navigation; **(B)** level selection; **(C)** quiz interface; **(D)** scoring and badge display. Licensed under CC BY-NC-ND 4.0.

elements (e.g., historical context, social insights) required for downstream applied systems.

Stage II (Active) comprises interconnected research spanning text processing, knowledge structuring, cross-language bridging, and cultural interpretation. Through these efforts, writing systems and their cultural assets evolve from passive digital archives into active resources supported by semantic organization, cross-linguistic readability, and contextualized explanation. This enables them to participate in digital ecosystems effectively, laying the foundation for future applications and transmission.

Stage III: APPLIED. Stage III (Applied) emphasizes use-oriented deployment, shifting focus from knowledge organization to functional systems and user experience in real-world scenarios. Works at this stage operationalize technical resources and knowledge structures from previous phases into products (e.g., education, art) to re-contextualize scripts within contemporary social practices. Fostering public engagement and intergenerational transmission through practical application, this stage comprises four sub-topics: System Platform Development, Educational Application Development, Interactive Experience Design, and Visual Arts & Design, accounting for 23 studies (18%) in our sample.

System Platform Development aims to integrate fragmented digital outputs into public services via three pathways. First, tool-based platforms provide foundational writing and communication functions, such as a cross-platform Lontara input system enabling mobile and desktop usage¹²⁰. Second, integrated applications combine AI to offer complex functionalities, exemplified by an OCR-based Lampung script recognition and translation system¹²¹. Third, community-centered platforms, such as the Sirijunga cultural memory platform designed to preserve traditions and strengthen communal ties¹²².

Educational Application Development targets practical teaching and learning contexts for script transmission, embedding endangered

scripts into actionable learning tasks and practice mechanisms through mobile delivery, gamification, and instructional design. The predominant approach leverages mobile gamification, incorporating interactive quizzes and game mechanics to lower learning barriers and engage younger generations. This has been applied to scripts including Javanese¹²³ (Fig. 9), Sundanese¹²⁴, Sasak^{125,126}, Baybayin^{127,128}, and Jawi¹²⁹. Furthermore, studies also combine application development with educational theories and cultural content, such as embedding apps in the STAD learning model¹³⁰ or the ADDIE model¹²⁵, and proposing strategies to strengthen cultural sustainability and digital literacy¹³¹. This signals a shift toward scientific, pedagogy-driven interventions. Beyond apps, researchers explore diverse teaching platforms to reach wider audiences, such as using MOOCs for Dongba script learning¹³² and developing computer-assisted tools for advanced learners¹³³. Additionally, research addresses technical challenges in script rendering to support teaching¹³⁴ and integrates cultural elements into modern user interfaces to enhance attractiveness¹³⁵.

Interactive Experience Design, beyond education, emphasizes stimulating public interest and active participation. Through interaction feedback, narrative, and related strategies, it shifts scripts from a relatively static mode of display toward an exploratory, dynamic experience, fostering emotional connection and cultural understanding. To achieve this goal, virtual reality (VR) emerges as a popular approach, with applications unfolding in two distinct directions. One focuses on the structural understanding of scripts, such as creating an interactive learning space where users can experience Dongba script formation principles¹³⁶. The other aims at immersive scene recreation, exemplified by Nüshu's social ecology recreated through role-play¹³⁷, and Indigenous constellation stories presented under virtual skies¹³⁸. Additionally, Cutting-edge explorations use AI for creative reflection, such as simulating the language creation process with 'AI Nüshu'. Researchers explore the complex relationship between power, cultural

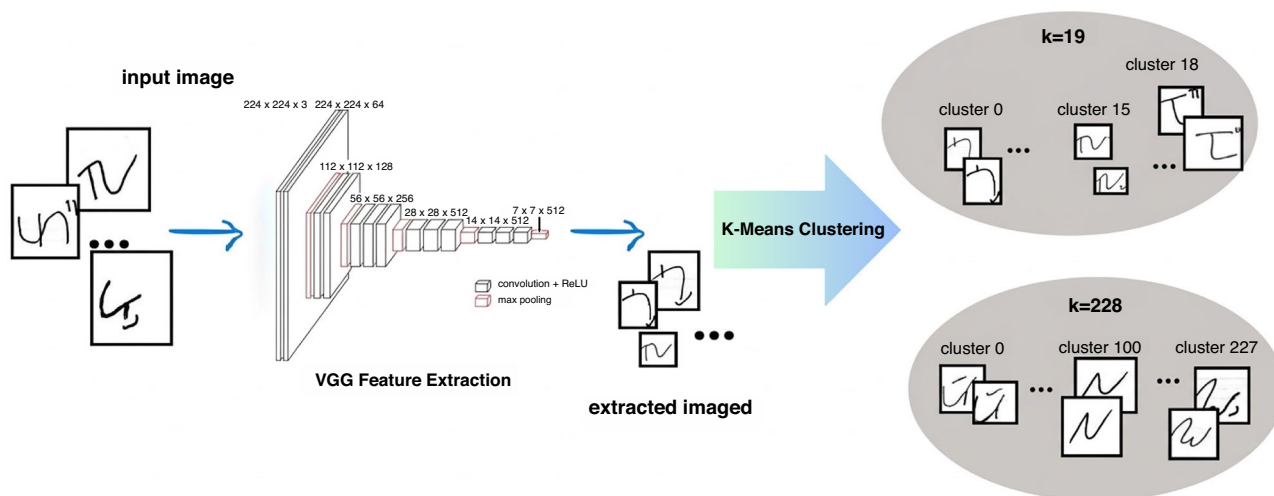


Fig. 10 | Workflow for image feature extraction and clustering of unannotated manuscripts⁵⁹. Images are processed with a VGG16-based network for feature extraction, then clustered by K-means ($k = 19, 228$) to group similar character samples. Licensed under CC BY 4.0.

heritage, and machine generation from the intersection of computational linguistics and feminism¹³⁹.

‘Visual Arts & Design’ primarily focus on visual representation and aesthetic communication, expanding the usage of endangered scripts from functional usage to compelling visual media. By combining social science insights with community participation, studies embed script-related cultural contexts into visual media, such as illustration, animation, and cultural creative products. The goal is to enhance public visibility and reach. Despite its potential to broaden dissemination, this category is underrepresented in our sample, with only one study identified: digital illustrations featuring the Japanese script¹⁴⁰.

The ‘Applied’ stage, through a series of practices, answers the core question of how endangered scripts can live in the present. This stage connects digital outcomes with societal vitality, from building functional platforms and educational applications to promoting interactive experiences and artistic expression. Whether endangered scripts’ digitalization can truly transition from technical survival to cultural revitalization depends primarily on the effective applications.

Technical tools & methods

Building on the 3A classification established in the previous section, this section reviews the major tools and methods employed to achieve the objectives of each stage. The aim is to construct a clear technical map that illustrates how existing technologies support research across the Alive, Active, and Applied stages, while also providing feasible pathways for future applications.

The technical toolkit of Stage I (Alive) establishes the foundation of ‘existence.’ In the technical pathway of ‘Dataset Construction,’ research practices show three distinct directions. (1) The ‘archaeological’ excavation of historical artifacts builds structured corpora from manuscripts and inscriptions through layout analysis, character segmentation, and manual verification⁴⁷. To improve efficiency, researchers integrate unsupervised clustering¹⁴¹, linguistic processing¹⁴², and web scraping-filtering strategies¹⁴³. (2) The ‘ethnographic’ approach captures living scripts through community-based collection, diversified sampling, and mobile applications⁴⁹, with semi-automatic annotation workflows^{46,48}, while hybrid strategies bridge domain gaps across datasets^{50,53,54}. (3) The ‘generative’ pathway employs GANs, DCGANs, and Diffusion Models to synthesize character samples, alleviating bottlenecks from scarce annotations and degraded historical materials^{14–16,43}.

‘Data Pre-processing’ ensures raw data quality through AI-driven restoration, applying generative models, such as GANs, Transformers¹⁵, and integrating them with professional software such as Adobe Photoshop⁵⁷.

Automation advances include CNN-based feature extraction with K-means clustering for rapid manuscript pre-structuring⁵⁹ (Fig. 10), and hybrid approaches combining classical algorithms with deep learning for human-AI collaborative processing⁶⁰.

Text Recognition (OCR), the central focus of Stage I, presents the richest technical approaches. Classical machine learning uses handcrafted features with traditional classifiers (KNN, SVM), remaining effective in small-sample scenarios^{73,144–147}. Deep learning introduced CNNs for recognizing individual characters or short words^{61,62,65,81}, with hybrid CNN-LSTM architectures handling continuous text⁸⁴, while Transformers address complex long-sequence dependencies^{44,49}. Another approach adapts object detection frameworks such as YOLO for word-level recognition⁵⁴. Furthermore, the latest study, DongbaBPN, employs GCN-based contour modeling to handle irregular pictographic shapes⁷⁷. Beyond large-dataset methods, recent few-shot architectures like dual-branch similarity networks improve adaptability under data scarcity⁵³ (Fig. 11), signaling progress toward frameworks that handle long sequences, irregular shapes, and limited data.

‘Annotation tool development’ advances through platformization. Mobile applications enable crowdsourcing⁸⁵, while modular collaborative platforms integrate AI with scholarly workflows⁶⁰. Font design and standardization create Unicode-compliant fonts, input methods, and keyboard applications to ensure script usability across mainstream platforms^{86–89}. In ‘Development Tools & Core Libraries,’ scholars expand existing libraries (e.g., Google’s Nisaba) via finite-state modeling⁹¹, or build Indigenous-language-centered programming systems with customized compilers and cross-lingual architectures⁹².

Stage II (Active) shifts to semantic ‘understanding.’ Technical tools at this stage concentrate on advanced analysis and structured processing, transforming Stage I outputs into searchable, linkable, and interpretable knowledge.

Knowledge formation begins with the computational deconstruction of text representation. ‘Natural Language Processing’ focuses on building orthographic and transliteration systems. For example, web applications and algorithms were developed for Balinese script to address complex scenarios ranging from derivational words⁹³ and affixes⁹⁵ to non-alphabetic symbols⁹⁴, mathematical expressions⁹⁹, and segmentation in scriptio continua^{97,98}. Building on this, research advanced toward language modeling and parsing. This includes improving word embeddings for complex languages¹⁰⁰, and transferring dependency parsing knowledge from mainstream languages to Xibe language¹⁰¹. To scarce data, studies explored graph-based models¹⁰², and semantic modeling to connect manuscripts with external knowledge bases¹⁰³.

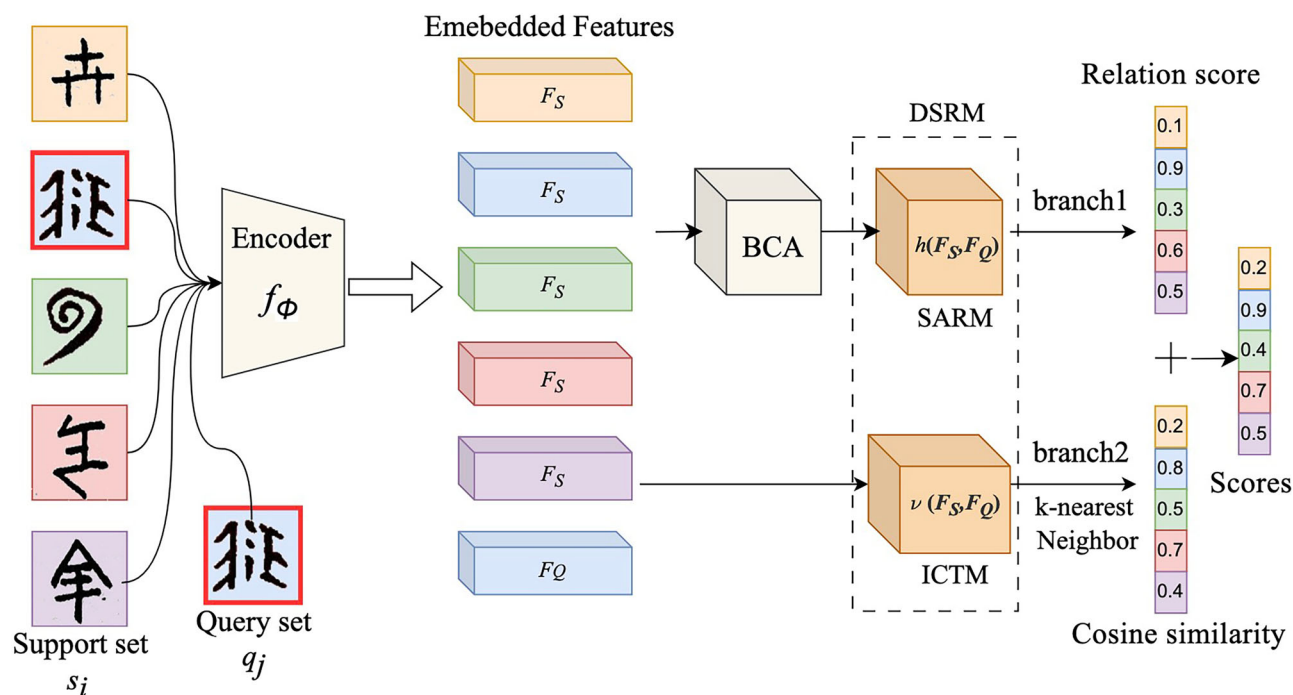


Fig. 11 | Overall architecture of the Dual-branch similarity network for few-shot Shui script recognition⁵³. The model uses an encoder for feature extraction and a Dual Similarity Relation Module combining Bi-Correlational Attention and k-

nearest neighbor similarity to improve recognition under limited data. Licensed under CC BY-NC-ND 4.0.

Once the internal structure of texts was parsed, research advanced toward two knowledge organization modes. On the one hand, ‘Knowledge Graphs Construction’ addresses data sparsity through joint models integrating multiple sources. Tibetan knowledge graphs combine graph structures with external textual information using deep learning for knowledge reasoning and querying¹³⁹. On the other hand, ‘Machine Translation’ establishes cross-language semantic channels through improving character handling and subword embeddings^{105,106}, transfer learning with external resources for Dongba script¹⁰⁷, and multimodal models with semantic augmentation¹⁰⁸. Additionally, end-to-end systems integrate computer vision, NLP, and linguistic rules to automate Nüshu manuscript translation¹⁰⁹.

‘Computational Humanities Research’ expands technology from computational methods to include social science and humanistic framework. At the computational level, researchers employ text and network analysis for data-driven interpretation^{110,111}, and web crawlers/APIs for community corpus collection¹¹⁸. More broadly, non-technical approaches include qualitative interviews and digital ethnography^{117,119}, semiotic analysis of historical scripts’ contemporary value^{112,114}, critical case studies of standards like Unicode¹¹⁵, and interdisciplinary frameworks integrating cultural research into type design^{113,116,148}.

The toolkits of Stage III (Applied) represents integrated practices that empower ‘application.’ Most technologies in Stage III build on those of Stage I and Stage II, characterized by a high degree of integration, with an emphasis on user experience.

‘System Platform Development’ and ‘Educational Application Development’ integrate Stages I and II techniques into unified mobile and web platforms for communities and public use. For example, an application embeds CNN-based character recognition to evaluate handwritten Baybayin and provide feedback¹²⁷, while others incorporate gamification for learning and cultural dissemination^{123,124,128,129}. Such development relies not only on UI/UX and image processing tools (e.g., Adobe Illustrator, Photoshop) but also on game engines (e.g., Construct 2) and backend services (e.g., Node.js, MySQL), ensuring a balance between functionality and user experience. ‘Interactive Experience Design’ practices are closely related to

media arts. Current efforts employ virtual reality and interactive narrative to create engaging experiences^{136–138}. ‘Visual Arts & Design’ emphasizes on cultural re-creation, clearly shown on digital illustration and visual design tools.

The above review of the three stages of technical tools reveals the current state of resources: a mature and well-developed Stage I set of foundational tools, a rapidly expanding Stage II set of analytical tools, and an emerging Stage III application integration scheme (Fig. 12).

Target Users

The ‘human’ element has always been at the heart of digitalization and revitalization of endangered scripts. A key question follows: who are these research outcomes for? Many projects pursue both technical advancement and public translation, though some focus primarily on one of the two. Existing studies often evaluate outcomes based on performance metrics (e.g., recognition accuracy, automation efficiency) or system usability. While these indicators effectively characterize technical progress, they struggle to fully address whether results enter the public sphere, gain comprehension among non-expert audiences, or foster actual practical usage. To examine the accessibility gap between experts and the public, we introduce Public Approachability as an interpretive analytical framework. We assess intended audiences qualitatively, based on output types, presentation forms, and user targets. Public Approachability is understood through three related dimensions: Technical Usability, Cognitive Friendliness, and Emotional Connection.

Technical Usability, grounded in usability engineering¹⁴⁹, examines whether an outcome presumes expert skills. Releasing results as code or software with complex configuration creates a high learnability barrier for non-experts. Cognitive Friendliness, aligned with cognitive load theory¹⁵⁰, evaluates whether content and interaction logic are designed for general learners through clear and consistent interfaces. Emotional Connection relates to principles of emotional design¹⁵¹. It assesses whether designs stimulate curiosity, foster emotional ties, and support cultural identity. Together, these three dimensions provide a holistic perspective on whether a research outcome extends beyond experts.

Stage	Research Objectives	Main Techniques
Stage I “ALIVE”	Give ES a Digital Identity displayed, recognized, retrieved, and reused.	• Generative Models (GAN, DCGAN, Diffusion, Transformer...) • CNN (VGG16, ResNet, CRNN...) • Sequence Models (LSTM, Transformer...) • Classical Machine Learning (SVM, KNN...) • Object Detection Frameworks (YOLOv8...) • Clustering & Feature Extraction (CNN+K-means, Edge Detection, Semantic Segmentation...) • Finite-State Modeling & Standardization (FST, Unicode Pipeline...)
Stage II “ACTIVE”	Make ES Semantically Active understandable, inferable, linkable, and translatable.	• NLP (Transformer, BERT, Graph-based Models...) • Knowledge Graph (CNN + Attention, Dual Embedding...) • Machine Translation (Subword Embedding, Transfer Learning, Multimodal OCR+MT...) • Cultural Research (Text Mining, Visualization, Digital Ethnography...)
Stage III “APPLIED”	Bring ES Back to Life used, learned, experienced, and revitalized.	• High Integration (Combining OCR, translation, NLP with Node.js, MySQL, Construct 2...) • User Orientation (UI/UX design, Adobe XD, Figma...) • Immersive Experience (VR, AR, GenAI, Affective computing...) • Creative Tools (Adobe Illustrator, Photoshop, Procreate...)

Fig. 12 | Research objectives and main techniques across the three stages of endangered script digitalization. The figure summarizes the corresponding objectives and techniques associated with the Alive, Active, and Applied stages.

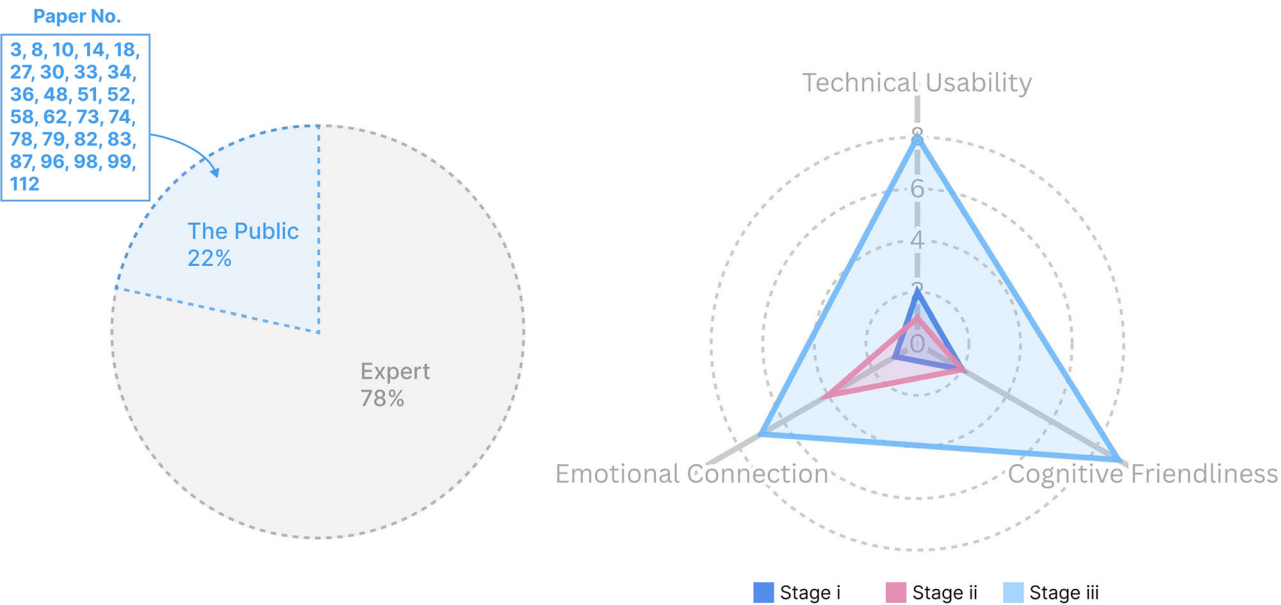


Fig. 13 | Audience orientation and public approachability of endangered scripts research. Left: distribution of target audiences, with most studies oriented toward experts (78%) and only a minority toward the public (22%). Right: evaluation of public approachability across three dimensions (Technical Usability, Cognitive Friendliness, Emotional Connection) at different research stages.

After categorizing and comparing 114 empirical studies, about 78% of the work targets experts (such as computer scientists, linguists, and historians), while 22% is oriented toward non-experts, mostly concentrated in Stage III (Fig. 13). Outputs in Stages I and II (such as datasets and models) are often released as data files, code repositories, or academic publications. They depend highly on professional skills, representing a typical “expert-serving-expert” model. Stage III studies directly target two categories of non-expert audiences. Educational applications lower learning barriers through mobile delivery and gamification, supporting community members, especially younger generations, in everyday literacy practice. Interactive designs, in turn, employ immersive settings, narrative content, and multimodal presentation to deepen understanding, engagement, and affective resonance

among the broader public. Together, these practices promote literacy learning and use, identity formation, and public dissemination.

Discussion

This section first summarizes the key findings and gaps revealed by the 3A classification, then examines how cultural grounding and community collaboration shape the validity and sustainability of digitalization. We finally propose future research directions for endangered script digitalization.

Key findings and gaps

Our systematic review of the past fifteen years, as presented in Sections 4 and 5, shows steady growth of research interest in endangered scripts

digitalization, along with significant technological advancement and maturation. Using the proposed 3A classification, the analysis also reveals the current status and existing gaps.

The literature shows a pronounced imbalance across the three stages. Stage I (ALIVE) dominates with approximately 60% of studies, aiming to make scripts computable and renderable. In contrast, Stage II (ACTIVE), focused on semantic organization and knowledge construction, and Stage III (APPLIED), oriented toward public dissemination and creative re-use, remain significantly underrepresented. This pattern indicates that the field is still in an accumulative phase, prioritizing technical assets (e.g., datasets, OCR models) over public dissemination and social impact mechanisms.

Steven Bird & Gary Simons emphasized the availability and accessibility of language resources¹⁵². Users should be able to access complete resources through clear and location-independent procedures. If research remains at the foundational level, digitalization risks a state of 'archiving for archiving's sake,' and fail to develop into an ecosystem that supports learning, dissemination, and reuse. Advances in emerging technologies offer unprecedented opportunities for digitizing endangered scripts, addressing unique challenges posed by scarce, fragmented, and degraded materials.

Our review shows that techniques, such as generative data augmentation, self-supervised learning, and transfer learning reduce label dependence and offer new solutions to persistent difficulties^{14–16}. These advances, however, are rarely delivered as complete and user-friendly systems, creating a significant gap between technical potential and practical value. In addition, the 'APPLIED' stage connects technologies to the public and has strong cultural influence, yet interdisciplinary approaches remain underutilized. Represented by 'Interactive Experience Design', approaches from media arts (such as multimodal sensory interaction, affective computing, et al.) offer rich creative possibilities for engaging the public and have shown substantial value in digital cultural heritage research²⁸, but remain underexplored in endangered scripts research.

Regarding target audiences, we have identified a distribution ranging from experts to the general public, but the coverage remains imbalanced. Over three-quarters of the outcomes primarily serve experts, while only a quarter offer high accessibility for language communities, learners, and the public. This review reveals that most studies were published in 2018 or later with a marked increase after 2020, many projects may still be transitioning from foundational technical and resource building toward broader impact translation. In particular, when researchers are still exploring and gradually clarifying which scientific outputs are suitable for conversion into public deliverables, it is not uncommon for outputs to concentrate on experts. Broader societal impact more often unfolds in later phases. Social scientists and designers are advancing creative translation and application development in parallel. UNESCO's *The Convention for the Safeguarding of the Intangible Cultural Heritage* emphasizes the importance of 'living heritage,' which lies in community practice, expression, and intergenerational transmission (<https://ich.unesco.org/en/convention>). Accordingly, language communities and the next generation are key actors in sustaining revitalization efforts, while broader public understanding, interest, and support also influence the dissemination effect, resource mobilization, and social recognition.

Overall, the major gaps in the practices of digitalizing endangered scripts point to barriers to a paradigm shift from Digital Preservation to Cultural Revitalization. The imbalance in target users skews the research focus, which further limits the openness of technical pathways and the development of innovative practices. High-accuracy digital capture and conversion form the foundation of digitalization, with objectives for encompassing creative and socially oriented contexts.

Cultural Context and Community Collaboration

Endangered script digitalization is best understood as a sociotechnical process rather than a purely technical workflow. Drawing on linguistic relativity⁶⁷ and the more recently proposed script relativity⁸, languages and scripts encode the cognition, worldviews, and broader cultural content of the communities. For example, pictographic systems like Dongba use spatial

arrangement and graphic composition to convey meaning, relying on visuospatial cues for comprehension¹⁰⁸. Nüshu practices are closely linked to women's social networks, reflecting female identity and affective expression¹⁵³. When endangered scripts are digitized, the preservation encompasses both the semiotic symbols and their underlying cultural contexts. This perspective has important implications for how we undertake digitalization work. Treating culture as an overarching framework that links the technical dimension with culturally specific meanings helps support project success. Within this process, the participation of Indigenous communities plays an irreplaceable role, as the value of participatory heritage is increasingly co-constructed through community interaction and multi-stakeholder engagement¹⁵⁴.

From the literature reviewed in this article, cultural knowledge permeates the three stages of digitalization. Community collaboration, contextual experience, and humanities and social science research methods influence whether subsequent technical and design work is valid, usable, and sustainable. **During the Alive stage**, the formation of data and resources relies heavily on community knowledge and governance structures. In terms of data sources, corpora typically derive from community-held ancient books^{33,141}, epigraphic inscriptions^{50,56}, printed texts⁵¹, or handwritten records^{48,49,59}, et al. Digitalization tasks such as training OCR/HTR models, building annotated corpora, and developing fonts/input methods require the community and research team to jointly define collection scope, public access, and authorization procedures. Lack of stable collaboration leads to data fragmentation, biased coverage, and cultural content mis-contextualization, undermining infrastructure reliability.

During the Active stage, many technical decisions are essentially modeling choices about cultural and pragmatic meanings. For example, the definitions of part-of-speech, entity boundaries, translation rules, and conceptual relationships in knowledge graphs all rely on context, pragmatics, and cultural categories. Accordingly, the emic perspectives offered by Indigenous communities can cross-validate the etic analysis of researchers¹⁵⁵. Taking 'Machine Translation' as an example, incorporating relevant cultural knowledge helps handle name-avoidance expressions in ritual contexts, preventing literal renderings that may cause misunderstanding.

At the Applied stage, projects generally target two non-expert user groups. The first is communities (user communities or origin/descendant communities), where research addresses local needs in language use, education, and transmission^{125,128,129}. The second is the broader public, where emphasizing accessibility and engagement stimulates external interest to enhance dissemination, thereby indirectly sustaining the heritage ecosystem^{136,138,140}. As scripts are tied to specific cultural knowledge, outputs influence how people understand and interact with them. For example, prioritizing aesthetics over context in public-oriented projects can reduce scripts to mere decoration, possibly leading to misinterpretation. In community-oriented education, learning a script also means understanding the cultural content and modes of thought that the Indigenous community carries¹⁵⁶. Neglecting these contextual rules hinders the transfer of literacy to real-world usage and undermines long-term transmission. Thus, collaboration with Indigenous communities is essential to maintain cultural integrity.

Historically, the vulnerability of languages and scripts stems not from natural decline, but from educational systems, religious discipline, and cultural assimilation under colonial rule^{3,26}. Therefore, although technology is indispensable to the digitalization process, it should not be the primary starting point. A more fundamental and critical element is the purpose of digitalization: preserving and transmitting cultural content using technical means. This requires the participation of Indigenous communities¹⁵⁷. Within collaborative frameworks such as CBPR^{31,32}, researchers can work with communities to make decisions about contextual interpretation/representation and to jointly define goals, content, and boundaries. Such considerations encompass which information is preserved or restricted, which narratives and translations are deemed appropriate, which symbols may be publicly displayed, and how cultural sensitivity, ownership, and

potential risks of misuse should be addressed. Effective collaboration between researchers and communities, with communities as equal partners at every stage³¹, ensures the clarity of objectives, boundaries, and success criteria, providing a solid foundation for subsequent technical routes and design forms.

Future directions

Building on our review and analysis of endangered script digitalization, we share our views on the future direction: sustaining technological innovation while systematically expanding cultural application and social engagement.

First, continued technical innovation in Stages I and II still remains crucial. Intrinsic script constraints, including small data scale, fragmented sources, heterogeneous forms, and limited semantic continuity, remain fundamental challenges in this field. Future work should advance low-resource learning and automated augmentation to alleviate data fragmentation and scarcity, and fill the synthetic-real transition gap. Furthermore, prioritizing systematic, long-term solutions, such as cross-script resource sharing and end-to-end data ecosystem development, to support sustainable reuse of technical outputs. Additionally, increased generative AI adoption necessitates integrating ethical governance, including data sovereignty, community consent, and content traceability. These issues concern research integrity and directly influence community acceptance and the long-term value of digital outcomes.

Second, future work should place greater emphasis on participatory approaches with communities, particularly Indigenous communities. Since scripts embody cultural contents, choices of technical pathways and output forms should be guided by goals, boundaries, and success criteria co-defined with communities. Consequently, digitalization processes can systematically adopt collaborative paradigms such as PAR/CBPR^{30,31}, embedding joint decision-making into problem framing, semantic annotation and interpretation, and dissemination strategies. Cultural research provides essential support for the paradigm shift from Digital Preservation to Cultural Revitalization, complementing the technology- and design-oriented analysis offered in this review. By recognizing both the achievements and the limits of current approaches, the field can advance toward a more holistic framework that balances computational innovation with cultural sensitivity.

Finally, our review shows that research on endangered scripts digitalization has been driven mainly by the computing discipline, with comparatively limited engagement from the design discipline. Yet, to transform preservation into 'living' cultural practices, the involvement of creative disciplines and cross-disciplinary collaboration is particularly critical. Trends such as immersive experiences, interaction design, and digital storytelling have emerged as new research hotspots²⁸. Open-ended artistic thinking and aesthetic expression¹⁵⁸, combined with the human-centered and iterative practices of design thinking¹⁵⁹, can help build richer cultural experiences. For script heritage in particular, however, design-driven work remains at an early stage. Accordingly, we propose the following priority directions.

- Systematic integration: Advancing user-oriented system integration by packaging mature Stages I and II technologies into complete, scalable, and accessible applications for non-experts.
- Human-centered design and community participation: Increasing alignment with community needs through human-centered and participatory methods, positioning communities as active cultural agents of cultural empowerment.
- Enhanced interactive experience: Supported by community collaboration and social science research, public-oriented projects can employ script-appropriate artistic and emerging interactive approaches (e.g., immersive interaction and digital storytelling) to enhance comprehension, engagement, and emotional resonance, moving beyond static display archives toward more participatory experiences.

Conclusions

This paper reviews the endangered scripts digitalization with a systematic classification. It maps the research landscape and contribution distribution, and identify core challenges. Findings show rapid growth over the last five years, a clear 'head-long tail' pattern on script types, and geographical concentration in Asia. Using the proposed 3A classification (Alive-Active-Applied), we illustrate the evolution from digital survival to cultural revitalization. Most studies (58%) focus on the Alive stage, establishing scripts' digital identity, while the Active stage (23%), and the Applied stage (18%) remain comparatively limited. It indicates that the field's research remains largely on the early stages of digitalization.

In recent years, the field has made notable progress, supported by increasingly mature technical tools and methods. Low-resource approaches, such as generative data augmentation, transfer learning, and few-shot recognition, have effectively addressed key challenges of endangered scripts (e.g., data scarcity, complex writing systems, data degradation, and high collection costs). Yet we still face three limitations: an imbalanced research focus, misaligned target audiences, and insufficient system integration. To overcome these challenges and enable a paradigm shift, future work should deepen technical innovation while strengthening cultural grounding and community engagement. It should also engage creative disciplines such as design, HCI, and the arts to advance community- and public-oriented applications and dissemination, supporting the transition from Digital Preservation to Cultural Revitalization.

To our knowledge, the survey is the first for endangered script digitalization in the last 15 years. Our 3A classification builds on the traditional technology-oriented taxonomy and further introduces a phased framework for analyzing the distribution of existing research. It offers a roadmap for newcomers and a structured lens to identify research gaps and future directions.

The survey, however, also has its limitations. First, this study takes Computing and Design as its primary analytical scope and reviews the literature from a technical and application-oriented perspective (research focus, computational methods, and user orientation). Although we briefly incorporate cultural context and community collaboration, broader humanities and social science issues are not discussed in depth. Future work could integrate these perspectives to develop a more comprehensive framework for assessing the cultural impacts and governance dimensions of digitalization. Second, our corpus is largely drawn from English-language academic publications, which may exclude community-led initiatives, regional documentation projects, and work published in other languages.

Future studies should develop methods to systematically incorporate diverse forms of knowledge production, including grassroots activities, local exhibitions, and institutional projects. These limitations further underscore that endangered script digitalization is a complex, long-term challenge that requires sustained interdisciplinary collaboration. We nevertheless believe that this review serves the audience interested or potentially interested in endangered script digitalization well, and may inspire future interdisciplinary collaboration between computing and design through its systematic and multidimensional analysis.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions

J.Q.S. wrote the first draft; F.T. and K.Z. added further discussions, comments, and proofread the manuscript. All authors have read and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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